

COLDPOD · DESIGN DECISIONS REGISTER

ColdPod design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coldpod		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

ColdPod design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes (pads and towers on the shell, rim slots, foam strip, loop in the can corner, block frame, liner feet and collar, folded lid tray, fixings, drip tray, cable duct)	(a) accept; (b) accept with changes	(a)	The whole build plan	CPD-DDR-003, A1
2	Mass (R12): 5.70 kg against 5.5 kg, now 0.20 kg over; replaces the earlier 0.03 kg question	(a) relax R12 to 5.75 kg; (b) keep 5.5 kg and weigh at TRL 4; (c) save mass now (1.0 mm tray, lighter handle and strap)	(a), since the added mass is the fixings the case needs; (c) if the target must stay	Lid tray gauge, handle choice	CPD-DDR-003, A2; CPD-DDR-002, N2
3	Refreeze (R8): all the PCM takes 8.0 h (8.03 h), a few minutes over 8 h	(a) restate R8 as 9 h for all the PCM; (b) accept the miss; (c) a thinner lid pack, trading passive hold	(a)	None in the build	CPD-DDR-002, N1; CPD-CAL-001 v0.4, G4
4	Battery removal (R14): the pack comes out by removing the battery bay with four screws	(a) count that as removable for the prototype; (b) add a shipping switch in the bay	(a) for the prototype	Battery bay wiring	CPD-DDR-003, A3
5	Working fluid and charge for the loop thermosiphon	Chosen with the refrigeration technician, for example R-134a or R-1234yf, by charge and local rules	Decide with the technician at TRL 4	Step 3 (charging)	CPD-DDR-003, A4
6	Co-design partner: an immunization program, a diabetes association or a humanitarian logistics group	Partner per area, chosen later	None yet	Not part of the TRL 3 build	CPD-DDR-001, O1
7	Appearance model: handle form, a window over the heat sink, the second status light, base bumper, labels and wordmark (docs/ REVIEW.md , 2026-09-26, items 1, 2, 5 and 6; items 3, 4 and 7 were adopted for construction in CPD-DDR-003, open under decision 1)	As listed in the review note	Round bail in the model; no window; second light adopted; labels under BOM line 16; bumper decided with mass (decision 2)	Renders and product model only	docs/ REVIEW.md , 2026-09-26

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The VIP maker can supply the seven panels to plus 0, minus 1 mm, with a quote and aged conductivity and edge data	The panels fill the shell exactly; joints and edges are about 45 % of the heat leak	CPD-DDR-003; CPD-CAL-001
2	The gasket still seals with the lid tray landed on the can flange, and the tray seat gives about 0.5 K/W	Lid seal and lid-pack refreeze both rest on it	CPD-DDR-002; CPD-CAL-001
3	The PCM maker can supply the eight pouch shapes, and the usable latent heat between 2 and 8 °C	Pouch fit round the risers and feet; hold times	CPD-DDR-003; CPD-CAL-001
4	The bought handle's pivot eyes take M5 screws and clear the 4 mm pads	Handle pivot pads	CPD-DDR-003
5	The off-state conductance and real COP of the chosen module with the chosen heat sink and fan	Hold power and the thermosiphon diode ratio	CPD-PRC-001
6	The Peltier driver can run the module from a 10 V input	R7 at the low end of the vehicle range	CPD-DDR-002
7	Whether a motorbike carrier stays within about 19° of tilt with the cooling head down; if not, the mechanical disconnect fallback	The loop thermosiphon works only below that tilt	CPD-DDR-001, D2; CPD-CAL-001, H5

■ Value engineering

Value-engineering target: USD 310 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 308 (USD 2 under the target).

- **Main cost drivers:** the vacuum-insulated panel set (about USD 60, still an estimate until quoted), the thermosiphon loop (about USD 28), the buck-boost Peltier driver (USD 22 to 26), the lid cold plate tray (USD 13) and the liner cut-out (USD 10 to 13).
- **Savings worth trying:** a VIP quote before any purchase; thinner printed parts, which already saved about USD 4 of filament; and re-pricing the driver and thermosiphon parts at purchase.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D11: 5 °C organic PCM, thermosiphon with a mechanical disconnect as fallback, evaporator on the outer face of the PCM, 76.8 Wh LiFePO4 pack, R6 redefined to 16 h, R12 relaxed to 5.5 kg, VIPs with a foam variant documented, local alarms and Bluetooth only, alarm defaults, budget \$300, outreach vaccinators first	Amish, going with the recommendation	CPD-DDR-001
2026-09-25	TRL 3 recommendations: lid cold plate, 3 °C liner cut-out, thinner printed parts, evaporator can and loop thermosiphon, TEC1-12703 class with smooth DC, buck-boost driver, budget kept at \$300	Amish: "i accept all your recommendations, go with them across all repos."	CPD-DDR-002
2026-09-26	Budget top-up to \$310 (R15 met)	Amish: "I am ok with the budget top ups"	CPD-DDR-002, N3
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The resulting changes are open for review (decision 1)	CPD-DDR-003